

# ABDELRAHMAN MAYASS HACHEM

Electrical Engineer | University Instructor | Embedded Systems, Automation & Robotics

Lebanon | +961 76 354 221 | [abdelrahman.hachem@ieee.org](mailto:abdelrahman.hachem@ieee.org) | [linkedin.com/in/abdel-rahman-h-b08bb781/](https://www.linkedin.com/in/abdel-rahman-h-b08bb781/)

## PROFESSIONAL PROFILE

Electrical engineer and university instructor with experience in embedded systems, control, automation, robotics, electronics prototyping, PCB design, 3D printing, and technical education. Experienced in teaching university-level engineering courses and developing hands-on projects using Arduino/AVR, ESP32, Raspberry Pi, MATLAB, LabVIEW, PLCs, and Python. Combines academic instruction with practical hardware development, simulation, troubleshooting, and project mentoring.

## CORE TECHNICAL SKILLS

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| <b>Embedded &amp; Robotics</b>  | Arduino/AVR, ESP8266/ESP32, Raspberry Pi, sensors, motors, GPIO, robotics integration                           |
| <b>Control &amp; Automation</b> | MATLAB, control-system modeling, PID, PLC automation, CodeSys, Twido, Zelio, PicoSoft                           |
| <b>Electronics &amp; PCB</b>    | EasyEDA, KiCad, Proteus, PCB Wizard, analog/digital simulation, circuit prototyping and debugging               |
| <b>Programming</b>              | Python, Arduino C/C++, MATLAB, HTML, CSS, JavaScript, Lua; Jupyter Notebook                                     |
| <b>3D Design &amp; Printing</b> | Fusion 360, Tinkercad, AutoCAD, Cura; Ender and Bambu Lab printers, slicing, maintenance and material selection |
| <b>Instrumentation</b>          | LabVIEW virtual instruments, optoelectronic sensors, electrical drawing and measurement systems                 |

## PROFESSIONAL EXPERIENCE

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| <b>Instructor</b>   Matn University College (MUC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mar 2024 - Present  |
| <ul style="list-style-type: none"><li>Teach Microcontrollers &amp; Microprocessors (AVR), Virtual Instruments with LabVIEW, Control Systems, Signals &amp; Systems, Optoelectronic Sensors &amp; Instrumentation, Electrical Drawing, PCB Design, and PLC Automation.</li><li>Deliver theory, simulation, laboratory work, and project-based instruction across electronics, embedded systems, control, and automation.</li></ul>                                                                                     |                     |
| <b>Coding &amp; Robotics Instructor</b>   Geek Express                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sep 2021 - Sep 2024 |
| <ul style="list-style-type: none"><li>Delivered courses in Arduino/Tinkercad, Micro:bit, VEX VR, web development, Scratch, and app development using MIT App Inventor and Thunkable.</li><li>Taught introductory AI and machine-learning workflows, including data collection, model training and evaluation, classification, and deployment concepts.</li><li>Guided learners through hands-on projects involving computer vision, sound/image classification, chatbots, and cloud-connected applications.</li></ul> |                     |
| <b>Physics Teacher</b>   Rooa Educational Compound, Tripoli                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Oct 2022 - Jul 2023 |
| <ul style="list-style-type: none"><li>Taught Grade 7 physics with emphasis on clear conceptual explanation and applied problem solving.</li></ul>                                                                                                                                                                                                                                                                                                                                                                     |                     |
| <b>Information Technology Teacher</b>   Rooa Educational Compound, Tripoli                                                                                                                                                                                                                                                                                                                                                                                                                                            | Oct 2021 - Jul 2023 |
| <ul style="list-style-type: none"><li>Taught Grades 4–11, including Microsoft Office, HTML/CSS, digital skills, and introductory programming concepts.</li></ul>                                                                                                                                                                                                                                                                                                                                                      |                     |
| <b>Freelance Engineering Project Advisor</b>   Independent                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2017 - Present      |
| <ul style="list-style-type: none"><li>Mentor senior and graduation projects through hardware design, simulation, testing, result analysis, technical documentation, and troubleshooting.</li></ul>                                                                                                                                                                                                                                                                                                                    |                     |
| <b>Robotics Instructor</b>   BLSC Center, Tripoli-Akkar                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Aug 2019 - 2020     |
| <ul style="list-style-type: none"><li>Taught block-based coding, hardware assembly, and robotics competition preparation for children.</li></ul>                                                                                                                                                                                                                                                                                                                                                                      |                     |
| <b>Advanced Robotics Instructor</b>   NCER Center, Tripoli                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Dec 2018 - Feb 2019 |
| <ul style="list-style-type: none"><li>Delivered university-level training in robotics design, DC/BLDC/servo/stepper motors and drivers, digital sensors, Arduino, and NodeMCU platforms.</li></ul>                                                                                                                                                                                                                                                                                                                    |                     |

## EARLIER EXPERIENCE

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| <b>Outreach Team Leader</b>   Safadi Accelerated Vocational Training, Safadi Institute                                                                                               | Nov 2017 - May 2018 |
| <b>Telecom Systems Technician</b>   Cell and Telecom Co., Tripoli <ul style="list-style-type: none"><li>Worked with telecom systems and VHF/UHF stations.</li></ul>                  | Nov 2014 - May 2015 |
| <b>Electrician &amp; Network Technician</b>   Techno New Co., Tripoli <ul style="list-style-type: none"><li>Worked on network systems and electrical wiring installations.</li></ul> | Apr 2014 - Sep 2014 |
| <b>Maintenance Intern</b>   Secure Vision Co., Tripoli <ul style="list-style-type: none"><li>Supported maintenance activities for CCTV security systems.</li></ul>                   | Jun 2013 - Aug 2013 |

## EDUCATION

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| <b>Master of Science in Electrical Engineering</b>   Lebanese International University                                                                                                                              | 2020 - 2022 |
| <b>TS1 in Electromechanics</b>   IPNET Technical Institute, Tripoli                                                                                                                                                 | 2020 - 2021 |
| <b>Bachelor of Science in Electrical Engineering</b>   Lebanese International University <ul style="list-style-type: none"><li>Senior Project: Self-Balancing Robot Based on PID Control - Grade: 95/100.</li></ul> | 2015 - 2020 |
| <b>Technical Bachelor in Electrotechnics</b>   Institute Technical Industrial - Tripoli (ITIT) <ul style="list-style-type: none"><li>Ranked 1st in North Lebanon and 10th in Lebanon, score: 1297/2000.</li></ul>   | 2014 - 2015 |

## SELECTED CERTIFICATIONS & ACHIEVEMENTS

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| <ul style="list-style-type: none"><li>Solar Technician Training - Greentech, November 2024.</li><li>Coding Educator Certification Program - CodeBrave, 2023.</li><li>Classification in Machine Learning - AI-iSolutions, February 2023: decision trees, random forests, SVM, logistic regression, model accuracy, analysis and visualization.</li><li>Clustering Techniques Training - AI-iSolutions, January 2023: K-means, mean shift, agglomerative clustering, DBSCAN and Gaussian mixture models.</li><li>Shortlisted by Tripoli Entrepreneur Club for the technology and safety startup “Gasofeel.”</li><li>Selected among seven projects from 33 submissions for the NAWAT Prize - Bomb Disposal Robot.</li><li>Participant, Build It Weekend 3.0 Hardware Hackathon, American University of Beirut, February 2019.</li></ul> |  |
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## LANGUAGES

**English:** Fluent | **French:** Academic/educational proficiency

## PROFESSIONAL INTERESTS & ACTIVITIES

Electronics and programming projects • Technical education • Robotics • 3D printing • Leadership and volunteering • IEEE Lebanon Section and IEEE LIU Student Branch activities • Beirut Mini Maker Faire